

AYUSH KUMAR SINGH

B.Tech CSE — Artificial Intelligence & Machine Learning

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SUMMARY

Computer Science and Engineering student specializing in Artificial Intelligence & Machine Learning, with hands-on experience building AI-powered web applications, browser extensions, data-driven dashboards, and full-stack projects. Interested in AI/ML, software development, intelligent systems, and practical problem solving.

EDUCATION

Narula Institute of Technology, Agarpara

B.Tech in Computer Science & Engineering — Artificial Intelligence & Machine Learning

Expected Graduation: 2029

TECHNICAL SKILLS

Languages: Python, C, JavaScript, HTML, CSS

Frontend: React, Vite, Tailwind CSS

Backend: FastAPI, Flask, Node.js, Express.js

AI/ML: Machine Learning, LLM APIs, AI Applications, Data Processing

Databases: PostgreSQL, SQLite, Firebase, Supabase

Tools: Git, GitHub, Docker, REST APIs, VS Code

Other: Chrome Extensions, Data Visualization, API Integration

PROJECTS

GhostAudit AI — AI-Powered Fraud Detection Dashboard

- Built a data-driven fraud analysis application for identifying suspicious transaction patterns from uploaded datasets.
- Developed a Python/Flask backend with data processing and AI-assisted analysis capabilities.
- Created an interactive dashboard with analytics and visualizations for investigating potentially fraudulent activity.
- **Tech:** Python, Flask, Pandas, NumPy, React, JavaScript, REST APIs, Docker

SatQuery AI — Remote Sensing AI Assistant

- Developing an AI assistant for interacting with and analyzing satellite imagery for Earth-observation use cases.
- Worked with satellite imagery, GeoTIFF data, and AI-based visual question answering workflows.
- Implemented a real raster-processing approach for bi-temporal satellite-image change detection.
- **Tech:** Python, AI/ML, GeoTIFF, Rasterio, Remote Sensing, VLMs

OrbitRakshak — Satellite Collision Intelligence

- Developed a decision-support concept for small-satellite operators to identify and understand potential orbital conjunction risks.
- Designed a workflow using public orbital data, numerical propagation, closest-approach analysis, and a risk assessment layer.
- Planned an interactive dashboard with what-if maneuver analysis and plain-language explanations.
- **Tech:** Python, FastAPI, SGP4/Skyfield, React, Vite, AI/LLM integration

VerifyX — Job & Internship Verification Platform

- Built a full-stack hackathon MVP focused on helping users verify job and internship opportunities.
- Developed a React frontend with a Node.js/Express backend and SQLite database.
- Added containerized deployment support through Docker Compose.
- **Tech:** React, Node.js, Express.js, SQLite, Docker

Chrome AI Summarizer

- Built a lightweight Chrome extension that summarizes webpage content using Chrome's built-in on-device AI Summarizer API.
- Designed the extension to process content locally, reducing the need for external server-side processing.
- Implemented the browser extension using standard web technologies and Chrome extension APIs.
- **Tech:** HTML, CSS, JavaScript, Chrome Built-in AI, Chrome Extensions

ACHIEVEMENTS & CERTIFICATIONS

- Participated in 2 hackathons and earned participation certificates.
- Built and published multiple AI, web development, and software projects on GitHub.
- Active project development in AI/ML, full-stack development, and emerging technologies.

PROFILES

GitHub: github.com/DevAyushTech

LinkedIn: linkedin.com/in/connect-ayushsingh/